CODEX HACKATHON | DESIGN DOCUMENT

Learning OS

A Codex-native study partner that makes you learn, not just collect answers.

In one sentence: Learning OS turns any material a learner already has into a gentle, citation-backed teach-and-check loop with automatic spaced review, built end to end on Codex's own features.

1 The Problem

AI makes people worse learners when it hands out finished answers, and better learners when it forces retrieval and reasoning. Two techniques have the strongest evidence for durable learning: **retrieval practice** (pulling an answer from memory instead of re-reading) and **spaced repetition** (revisiting material at growing intervals). In a semester-long study, students who actively used an AI tutor for personalized retrieval practice gained up to 15 percentile points.

Almost nobody studies this way alone, because doing it by hand is tedious: sourcing material, writing quiz questions, tracking confidence against actual accuracy, and remembering to come back on a growing schedule.

The gap: turning passive AI use into active learning, automatically, without relying on the learner's willpower to do the hard parts.

Who it is for

Self-directed, independent learners studying outside a formal classroom structure: adults picking up a new skill, people preparing for certifications, and university students studying material their course does not regularly quiz them on. The system is subject-agnostic, so chemistry, music theory or a coding framework all work the same way.

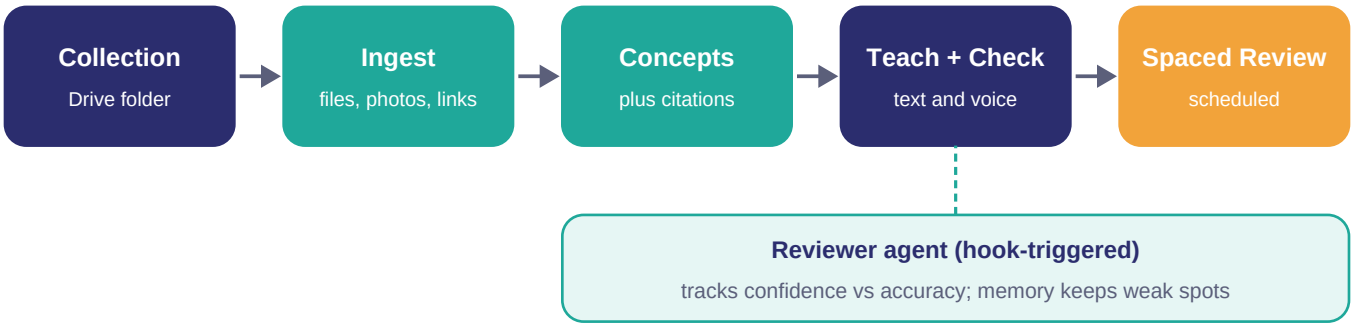
2 Core Design Principle

Teach, then check, in small loops - never teach everything and then exam everything.

Layer	What happens
Within a session	One concept at a time, grounded in the learner's own material, followed immediately by one question or an "explain it back" prompt. The check feels like confirmation, not a separate test.
Across sessions	A scheduler resurfaces older material at increasing intervals. This layer is more purely retrieval, since the goal is to bring it back before it fades.
Throughout	The system tracks the gap between how confident the learner feels and how they actually perform (confidence calibration, a form of metacognition).

The tone is a study partner that notices what is shaky and casually resurfaces it, not an exam.

3 How It Works at a Glance



4 Ingestion Pipeline

Material arrives **incrementally**: a chapter, a lecture deck, or a session's worth of notes at a time, never one giant upload.

Source	How it is handled
PDFs	Textbook chapters, slide decks, past questions. Page numbers kept for citation.
Typed notes	Plain text and markdown.
Photos	Pages or handwriting snapped on a phone and read through Codex's native image input.
Spreadsheets	Reference sheets. Cell references kept for citation.
Web pages	Public articles pulled in through MCP from a pasted link.
YouTube links	Transcript or captions fetched and treated as text, with timestamps kept. Codex does not watch the video.
Stretch goal	Audio or video file transcription. Not core scope.

From material to concepts

Each chunk is broken into discrete, testable **concepts** rather than treated as one blob of text. As chapters accumulate, the system builds a growing knowledge map, so later material can be quizzed against earlier material, not only in isolation.

Source-grounded citations

Every concept keeps a pointer to exactly where it came from: page, slide, transcript timestamp or spreadsheet cell. The citation travels with every explanation and question, so asking "where is that written?" never triggers a fresh search, because the origin was tagged at ingestion.

- **PDFs:** jump to and highlight the originating page where feasible.
- **Other formats:** quote the precise location (cell reference, timestamp) instead of highlighting.

Scope boundaries

In scope	Out of scope
Files, photos, public links, and the learner's own connected Drive and Gmail.	Gated third-party content such as a paywalled Udemy course. The architecture is extensible through MCP, but nothing unbuilt is claimed as working in the demo.

5 UI Layout: Collections

The organizing unit is the **Collection**, a folder-like grouping the learner defines. Each Collection maps to a Google Drive folder through MCP, so uploading is simply dropping files into Drive. Ingestion, quizzing and spaced review are all scoped per Collection, and each one keeps extending its own knowledge map over time.

Personal Development after-school and self-training	**University Courses** school material	**Musical Training** practice and theory

6 MCP and Google Drive Access

- **Google Drive:** backs the Collections model. New files dropped into a folder are picked up and ingested without a fresh manual upload.
- **Gmail:** available where source material or reminders flow through email.
- **Web and URL fetch:** pulls public pages, articles and YouTube transcripts into the same pipeline.

> **How MCP works.** An MCP server exposes named tools (for example search_messages or get_file_content) to Codex. Codex decides which tool to call and when. The server holds the authenticated connection, such as an OAuth token, makes the real API call, and returns a clean result. The protocol is universal; each tool is only as capable as the wrapper built around the service. For services with no official MCP server, a generic browser-automation MCP such as Playwright MCP is the realistic path, noted architecturally but not built here.

7 Codex Features Exercised

Codex feature	Role in Learning OS
Subagents	Planner, Quizzer and Reviewer run as distinct agents. Quizzers run in parallel across topics.
Git worktrees	Isolate parallel Quizzer runs per topic or Collection so they never interfere.
MCP	Drive and Gmail access, plus web fetch for links and YouTube transcripts.
Skills	"Generate a retrieval-practice set from ingested material" packaged as a reusable skill.
Hooks	The Reviewer fires automatically when a teach-and-check loop or quiz session ends.
Scheduled tasks	Native recurring automation re-triggers spaced-repetition reviews at growing intervals instead of faking it in custom code.
Memory	Confidence history and per-Collection knowledge maps persist across sessions.
Image inputs	Photos of notes and pages are ingested natively.
Voice	Conversational teach-back and check-ins.

8 Interaction Media

Three channels work together rather than one being chosen exclusively.

TEXT	VOICE	IMAGE
The backbone. Written explanations with citations, quiz questions, and a review of history.	Genuinely two-way and turn-based, not record-and-submit. The learner explains a concept aloud, the system asks a Socratic follow-up on the weak spot, and they go back and forth within one session. Turn-based, not overlapping real-time dialogue.	Two-way. Photos of notes and pages come in; a highlighted source page or diagram goes out where the format allows.

9 The Experience, End to End

A learner creates a Collection and drops material into its Drive folder in whatever form they have: a PDF, a photo, a link. The system ingests it incrementally, extracting citable concepts and building a growing knowledge map. In a session it teaches one concept at a time from that material, checks it conversationally (often by voice), and tracks confidence against accuracy. Across days and weeks, a scheduled automation brings older material back at growing intervals. Asking where something came from always returns a precise citation.

The result: passive content becomes an active, low-stress learning loop, run largely by Codex's own native features rather than custom-built infrastructure.

10 Selective Scope and Progress Tracking

Choosing what counts as the syllabus

Ingestion does not automatically turn an entire uploaded chapter into required material. Once a material is ingested, it is broken down into its titled sections, for example Chapter 3 and Chapter 4 each surface their own list of titles, shown side by side regardless of which material they came from. The learner selects exactly which titles to work on now, mixing and matching across multiple materials if they choose. Whatever is selected becomes the 100 percent, the denominator the learner is actually measured against, not the full content of everything uploaded.

Reordering

Materials and titles can be re-sequenced by drag and drop. If something was uploaded out of order, the learner can drag it to where it belongs, so study order is set by the learner, not by upload order.

Progress tracking

A visible progress indicator reflects how much of the learner's self-selected scope is complete, giving them something to return to and check against.

Marking a title complete: AI-judged, confidence-aware

Completion is not a fixed rule such as one correct answer or three in a row. The Reviewer agent owns this judgment using the same confidence-calibration signal from Section 2: a title is marked complete once the learner answers correctly and confidently, with no sign of guessing or hesitation. The bar is good enough, move on; the Reviewer does not over-test a title once it is satisfied.